

Roll Number: _____	
Thapar Institute of Engineering and Technology, Patiala	
Computer Science and Engineering Department	
Mid-Semester Examination	
BE COE & CSE (Third Year)	UCS701: Theory of Computation
21 st March 2024, 11:00AM	Time: 2Hours, Max Marks:30
Instructors: Sunita Garhwal, Nidhi Kalra, Shashank Sheshar, Aditi Sharma, Gourav Jain, Deepika Shukla	

Note: Attempt all questions. Assume missing data, if, any, suitably.

Q No.	Questions	Marks	CO	BL
Q1	Given the regular expression $r=ab(a+b)^*ba$ over the alphabet $\Sigma=\{a,b\}$ a) Convert the given r into NFA using Thompson's construction. b) Convert the obtained NFA into DFA using subset construction. c) Minimize the obtained DFA in 1 (b).	2 2 2	CO1	L3
Q2	Given $L_1 = \{x \mid x \text{ ends with } 10, x \in (0,1)^*\}$ $L_2 = \{x \mid 10 \text{ is not part of string of } x, x \in (0,1)^*\}$ Draw the Finite Automata recognizing the following languages: a) $L_1 \cap L_2$ b) $L_1 \cup L_2$ c) $L_1 - L_2$	6	CO1	L4
Q3	a) Design a Mealy machine that generates output 1 if occurrence of 'a' and 'b' are both are even, output 2 if occurrence of 'a' is odd but occurrence of 'b' is even, output 3 if occurrence of 'a' is even but 'b' is odd, and output should be 4 otherwise. b) Convert the following Moore Machine to Mealy machine. c) Find all the strings in the language $L=(abc^*de)$ of length less than eight.	3 2 1	CO1	L3 L4 L2

Q4	a) Let L be the language over $\Sigma = \{0,1,2\}$ consist of strings that do not have any consecutive 0's, consecutive 1's, or any consecutive 2s. Is L regular? If yes, then design NFA for L else prove using Pumping lemma that L is not regular. (00,11,22 are few examples of Rejected strings whereas 01,12,02 are few examples of Accepted strings.) b) Using Pumping Lemma, Prove that $L = \{w \in \Sigma^* \mid n_a(w) > n_b(w)\}$ over alphabet $\Sigma = \{a,b\}$ is not regular.	3 3	CO2	L4 L4														
Q5	a) Generate Regular expression for the following automata using State Elimination Method. <pre>graph LR; Start(()) --> S(((S))); S -- c --> S; S -- a --> F(((F))); F -- b --> S; A((A)) -- a --> A; A -- c --> F;</pre> b) Write the right linear grammar for the language $\Sigma = \{a,b,c\}$ which contains one or more a's followed by one or more b's followed by one or more c's. c) Convert the following NDFA into DFA over $\Sigma = \{a,b\}$ using table method: <table><thead><tr><th rowspan="2">Present State</th><th colspan="2">Next state</th></tr><tr><th>a</th><th>b</th></tr></thead><tbody><tr><td>→A (initial state)</td><td>A, B</td><td>C</td></tr><tr><td>B</td><td>A</td><td>B</td></tr><tr><td>C (final state)</td><td>B</td><td>A, B</td></tr></tbody></table>	Present State	Next state		a	b	→A (initial state)	A, B	C	B	A	B	C (final state)	B	A, B	3 1.5 1.5	CO1	L3 L3 L4
Present State	Next state																	
	a	b																
→A (initial state)	A, B	C																
B	A	B																
C (final state)	B	A, B																

Marks Distribution

